Ahmed Rockey Saikia

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Gender: Male
Date of Birth: 30th October 2004
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Nationality Indian
Permit B (Student)

Education

Sep 2025 – ongoing **M.S. in Computational Science and Engineering**, *EPFL, Switzerland*
GPA: 5.4/6 Relevant Courses: Systems for data management and data science, Maths of Data:Theory to Computation, Parallel and High Performance Computing, Machine Learning, Deep learning, Stochastic Calculus

Nov 2021 – May 2025 **Bachelor of Technology in Engineering Physics**, *Indian Institute of Technology(**IIT**), Roorkee*
GPA: 9.3/10 (Department Gold Medalist)
Relevant Coursework: Machine Learning and Deep Learning, Data Structures, Algorithms, Numerical Analysis

Experience

Feb 2026 – ongoing **Student Research Assistant**, *LIONS Lab, EPFL*, Switzerland
Design and develop a distributed machine learning framework to train Diffusion-LLM using PyTorch DDP, Hydra and WandB to implement novel discrete diffusion architectures through masked and uniform noise schedules.

Jan 2026 – ongoing **ML Research Assistant**, *Swiss AI Initiative, ETH & EPFL AI Center*, Switzerland
Developed a multi-node pipeline for audio transcription, speaker diarization and annotation using vLLM to preprocess around 100k hours of audio data. Also, integrating a dynamic frame rate tokenizer into the Apertus LLM for interleaved audio-text token for multimodal pretraining.

July 2023 – Aug 2023 **TSSP Fellow**, *Nicolaus Copernicus University*, Torun, Poland
Developed an image stitching pipeline using Python and OpenCV to process high resolution biomedical OCT images to generate wide-field composites via SIFT and RANSAC homography estimation to align image sequences.

May 2023 – July 2023 **SPARK Fellow**, *Indian Institute of Technology, Roorkee*, India
Developed a distributed C++ simulation framework using OpenMPI to manage inter-process communication for large scale lattice computations across multi-node cluster environments. [DOI: 10.1103/PhysRevB.110.134430]

Projects

Nov 2025 **Neural Audio Codec** - Implemented residual vector quantization based on the Soundstream architecture to compress signals into discrete low dimensional latent embeddings for high-fidelity reconstruction.

Oct 2025 **Muon and Dion Optimizer Benchmarking** – Benchmarked Muon and Dion optimizers, implemented using PyTorch, against Adam for training ResNet and Vision transformers on the CIFAR-10 dataset.

Awards & Hackathons

Nov 2025 **ApartResearch defensive Hackathon**, Collaborated with a diverse 5-person team to develop a CNN based ML model that detects sensitive genomic sequences and mitigates adversarial inputs.

Oct 2025 **3rd prize in Argusa AI challenge**, Ranked 3rd among 20 competing teams by developing a multimodal Retrieval-Augmented Generation(RAG) application with LlamaIndex, Docling and Streamlit.

Sep 2025 **Professional Development and Innovation Award**, Awarded for overall excellence at IIT-R

May 2024 **DAAD-WISE Fellowship**, Selected from across India to conduct research at a German university (TUM).

Jan 2024 **UG Chanakya Fellowship**, Awarded by iHUB, IIT Roorkee to conduct a student project in AI/ML.

Technical and Soft Skills

Programming Languages Python, C/C++, Scala, SQL
Tools and Frameworks PyTorch, Docker, Hugging Face, Langchain, Hydra, Apache Spark, Numpy, SciPy, Git, Bash
Languages English (Advanced), Hindi (Advanced), Assamese (Native)

Extra-curricular Activities

Mar 2024 – Mar 2025 **Vice-President**, *Physics Association, IIT Roorkee*, India
Organized departmental events and facilitated student–alumni interactions to support student career guidance.